Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/08/2026
PI Name:	Ling, Yue		
Project Title:	Multi-Scale Simulations of Two-Phase Flows and Heat Transfer in Confined Geometries Using Embedded Thin-Film Models		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
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Zip Code:	29208-4104	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-PSI E.8 Physical Sciences Informatics
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2026 per J. McQuillen/GRC (Ed., 9/29/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tryggvason, Gretar Ph.D. (Johns Hopkins University)		
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Task Description:	Multiphase flow and heat transfer in confined geometries are important, but challenging to simulate. In the microgravity environment, confined two-phase flows often form thin liquid films, separating the interface from the wall. The thin liquid films place a fundamental multi-scale challenge in numerical simulation, since a direct simulation that resolves both the large scales and the thin films is exceedingly expensive. This project aims at establishing a multi-scale model which couples the subgrid thin-film models with the interface-resolved front-tracking method, to tackle this challenge. The front-tracking method represents the interface by a separate surface grid ("front"), which makes it convenient to embed models on the interface, compared to other interface-capturing methods. The Capillary Channel Flow (CCF) CCF-PS-EU2 experimental results, including the bubble position evolution and the rate of phase separation, are critical to the validation of the proposed multi-scale model in capturing the wall and free films and their impact on bubble migration and bubble-surface (or bubble-bubble) interaction. The proposed model is important to the design and performance prediction of multiphase flows and thermal systems of interest to NASA. The model will be implemented in the open-source multiphase flow solver PARIS (PArallel, Robust, Interface Simulator), developed and freely distributed by the Principal Investigators (PIs), to make it available to the multiphase flow community. The proposed multi-scale framework is capable of capturing heat transfer and a thermocapillary effect that is not currently considered in the CCF experiment and we expect the outcomes will also inspire new experiments in the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Multiphase flow and heat transfer in confined geometries are important but challenging to simulate, as a direct simulation that resolves both the large scales and the thin films is exceedingly expensive. The proposed model will enable efficient and accurate simulations of such flows. The proposed simulation results will inspire new experiments in the future.
Task Progress:	During this reporting period, progress was made toward the development, validation, and application of a computational framework for simulating confined bubble migration and breakup in wedge-shaped geometries under microgravity conditions. The work builds on prior efforts to address the challenges associated with resolving multiscale interfacial phenomena, particularly the interaction between large-scale bubble motion and thin liquid films that form near solid boundaries. Because directly resolving these thin films is computationally prohibitive, the current study continues to employ a modeling strategy that captures their effects through a physically motivated wall interaction model. This approach enables accurate and efficient simulations of bubble migration, deformation, and breakup in confined geometries. A central effort during this period involved conducting a systematic set of high-fidelity three-dimensional simulations to investigate how physical parameters influence bubble behavior after gravity is removed. The study focused on the effects of initial Bond number, wedge angle, and liquid viscosity, which together determine the initial shape of the bubble and the forces governing its subsequent motion. These simulations were designed to replicate conditions from microgravity experiments, allowing for direct comparison and validation. The results show that the initial shape of the bubble, established while gravity is present, plays a dominant role in determining its later evolution. Although gravity is removed during the primary stage of interest, its influence persists through the initial deformation of the bubble. Bubbles that are more elongated at the time of release are more likely to undergo significant deformation, develop thin necks, and eventually break apart. In contrast, more compact bubbles tend to remain intact as they migrate. Across the range of conditions examined, three distinct types of behavior were observed: cases in which the bubble remains intact, cases in which it breaks into two smaller bubbles, and cases in which it breaks into three bubbles. These different outcomes are primarily determined by the degree of initial deformation and the balance between capillary forces and viscous effects. As the initial deformation increases, the system transitions from no breakup to more complex fragmentation patterns. The geometry of the system, represented by the wedge angle, was found to have a strong influence on these behaviors. Smaller wedge angles lead to greater confinement and more elongated bubble shapes, increasing the likelihood of breakup. Larger wedge angles, by contrast, produce more compact bubbles and tend to suppress breakup. In addition to influencing whether breakup occurs, the geometry also affects how the bubble deforms and how any resulting fragments move. Liquid viscosity plays an equally important role by damping fluid motion and reducing the extent of deformation. Higher viscosity tends to smooth out rapid changes in bubble shape, delay or prevent breakup, and reduce oscillations during migration. In some cases, increasing viscosity was found to change the outcome entirely, preventing breakup that would otherwise occur under lower-viscosity conditions. These findings highlight the importance of viscous effects in controlling bubble dynamics in confined environments. The computational results were compared extensively with available experimental data from microgravity studies. The simulations show good agreement with experiments in terms of bubble shape evolution, breakup behavior, and overall migration trends. This agreement is particularly strong during the early stages of motion, when capillary forces dominate and the key physical mechanisms are most clearly observed. The consistency between simulations and experiments provides strong validation of the modeling approach and supports its use for predictive studies. An important outcome of this work is the development of a regime map that summarizes how different combinations of parameters lead to different types of bubble behavior. This map identifies regions corresponding to no breakup, breakup into two bubbles, and breakup into three bubbles. It shows clear trends indicating that increased deformation promotes breakup, while increased viscosity suppresses it. The regime map provides a useful framework for understanding the combined effects of physical parameters and for guiding future experimental and modeling efforts. In addition to identifying overall trends, the simulations also provided insight into the motion of bubbles after breakup. In cases where multiple bubbles are formed, the fragments can interact with each other and with the surrounding fluid in complex ways, including moving toward each other or remaining separated depending on the conditions. While these detailed dynamics are not the primary focus of the current report, they demonstrate the capability of the computational framework to capture a wide range of physical behaviors.

	The results obtained during this reporting period have been disseminated through a peer-reviewed journal publication and a conference presentation. These efforts contribute to the broader scientific community and help establish the validity and relevance of the work. Additional results from the current study are being prepared for future publication. Overall, this work represents meaningful progress toward the project's goal of developing a reliable and efficient modeling framework for confined multiphase flows. The combination of systematic simulations, validation against experiments, and the development of generalized trends provides both improved physical understanding and practical tools for predicting bubble behavior in microgravity environments. These advances lay a strong foundation for continued research in the next reporting period.
Bibliography Type:	Description: (Last Updated: 05/26/2026)
Abstracts for Journals and Proceedings	Ahmed S, Tryggvason G, Ling Y. "Rise of a confined bubble in a wedge channel under different gravity." APS Division of Fluid Dynamics Annual Meeting, Houston, TX, Nov. 23-25, 2025. Abstracts. APS Division of Fluid Dynamics Annual Meeting, Houston, TX, Nov. 23-25, 2025. , Nov-2025
Articles in Peer-reviewed Journals	Ahmed S, Tryggvason G, Ling Y. "Rise of a confined bubble in a wedge." Int. J. Multiph. Flow. 2026 Jan 22:105628. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2026.105628 , Jan-2026